

2022-Ph-H1-Q15

Section: Our Dynamic Universe

Topic: Measure, Gravitation, Relativity

Which observation supports the idea that time runs more slowly in stronger gravitational fields?

Answer: E

(Time passes more slowly near the surface of a neutron star than far away from it.)

Guidance for Students:

This is **gravitational time dilation**, predicted by general relativity: time runs slower in stronger gravitational fields.

Revision Tips:

- Closer to massive objects → slower time.
 - GPS satellites correct for time dilation.
 - Neutron stars have extremely strong gravity.
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